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Magnetostratigraphy of the Miocene–Pleistocene Surai Khola Siwaliks in West Nepal

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SUMMARY

Magnetostratigraphic age dating has been carried out on sandstones of a well-exposed continuous 5500 m thick Miocene–Pleistocene section in the Surai Khola area (West Nepal). The section is situated in the Siwalik Range which consists of molasse sediments from the Himalaya. Progressive demagnetization and isothermal remanence experiments prove that haematite is the carrier of the characteristic remanent magnetization (ChRM) throughout the entire section. The final polarity sequence is based on 436 samples, of which 233 have normal and 203 reverse polarity. Antiparallel mean directions of the normally and reversely polarized samples indicate that the ChRM is of primary origin. Comparison of the polarity sequence with a standard polarity time-scale yields a time period for the section between chron 1r or chron 2r (Matuyama) and probably chron 5AB (between 1–2 Myr and about 13 Myr).

Key words: Himalaya, magnetostratigraphy, Neogene Siwaliks.

1 INTRODUCTION

The Siwalik Range is formed by Miocene to Pleistocene molasse sediments to the south of the Himalayas.

Since 1983 lithological, biostratigraphic and archaeological studies have been in progress in the Surai Khola area of West Nepal (Fig. 1) (Corvinus 1988). This study is concerned with a continuous, well-exposed, sedimentary section of about 5500 m thickness. The section consists of 60°–80° northward dipping and WNW–ESE striking sediments without any major tectonic disturbances. It represents continuous fluvial sedimentation during the Neogene period. The section is composed of interbedded conglomerates (mainly in the upper part), sandstones, clays, shales and siltstones (Fig. 3). The grain size of the sandstones coarsens from the lower to the upper part of the section. A lithostratigraphic subdivision into six units (Bankas, Waterspring, Chor Khola, Surai Khola, Dobatta and Dhan Khola Beds) has been proposed by Corvinus (1988).

Detailed biostratigraphic age dating seems to be impossible, and radiometrically datable volcanic ashes have not yet been detected. Successful application of magnetostratigraphy in other areas of the Siwalik Range (Opdyke *et al.* 1979; Johnson *et al.* 1982; Tauxe & Opdyke 1982; Johnson *et al.* 1983; Burbank & Johnson 1983; Tokuoka *et al.* 1986) encouraged us to apply this method in the Surai Khola section. Preliminary results have been reported by Appel *et al.* (1989).

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ANALYSIS OF REMANENCE COMPONENTS AND ROCK MAGNETIC STUDIES

Oriented sandstone samples (594) were drilled across the entire section. Only 20 are from the mainly conglomeratic Dhan Khola Beds at the top of the section. In order to obtain a better resolution, 74 additional oriented clay samples were collected there. However, the clay samples yield no consistent behaviour. This paper is therefore concerned with sandstone samples only.

Despite lithological differences (e.g. grain size and colour) of sandstones within the section, the demagnetization behaviour is quite homogeneous. The characteristic remanent magnetization (ChRM) is obtained from suitable stepwise thermal demagnetization suggested by pilot sample studies (AF demagnetization not adequate to separate hard magnetic haematite and goethite remanences). For the thermal treatment shielded SCHONSTEDT or MMTDM furnaces were used. Susceptibility measurements after each step of heating enabled a control of possible mineral alterations. Remanence measurements were carried out with a cryogenic magnetometer (LETI RS 01). Generally NRM intensities (mostly between 0.1 and 10 mA m^{-1}; see Fig. 3) and remanence intensities after suitable steps of demag-

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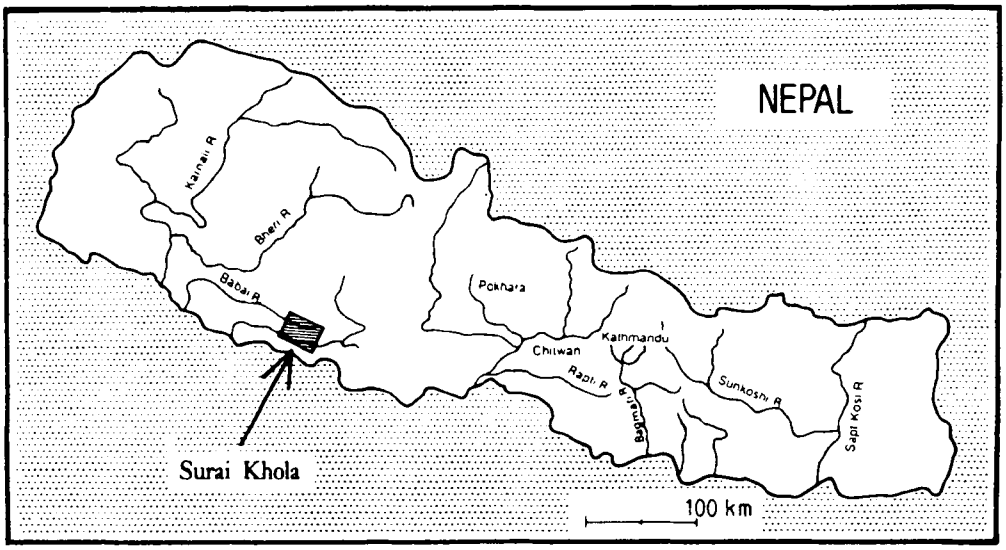


Figure 1. Map of Nepal with investigated area of Surai Khola.

netization were usually well above the noise level (about 0.01 mA m^{-1}) of the magnetometer.

Isothermal remanent magnetization (IRM) acquisition and thermal demagnetization of an IRM and of the natural remanent magnetization (NRM) were used to identify the ferrimagnetic minerals by the critical parameters coercivity (H_C and blocking temperature (T_B) (Fig. 2). Haematite is observed in all samples (high H_C and T_B around 650°–700 °C). Many samples contain also a varying amount of goethite (high H_C and T_B of about 100 °C; e.g. samples 74–99, 48–1 and 46–3 in Fig. 2). In some of the samples magnetite (or maghaemite) also occurs (lower H_C and T_B around 600 °C; e.g. samples 245–1 and 48–1 in Fig. 2). The goethite content is higher in the lower part of the section, as indicated by the strong decay of the remanence intensity at 200 °C [ratio dJ(TH200)/J(TH200) shown in Fig. 3]. Magnetite is negligible below 1800 m, dominates between 3000 and 3300 m, and is of intermediate importance in the rest of the section. This is shown by the ratio IRM(0.21 T/1.5 T) in Fig. 3. Magnetite is almost saturated at 0.21 T, whereas saturation of haematite requires about 10 times higher fields. Hence magnetite-rich samples are identified by values close to 1.

Samples with haematite or with haematite and goethite show a single remanence component upon thermal demagnetization from 100 °C up to about the Curie temperature of haematite (e.g. 74–99 and 46–3 in Fig. 2). This component is accepted as the ChRM direction. Chemical remanences carried by goethite (parallel to the present field direction) are completely destroyed at a temperature of 100 °C. Viscous remanent magnetizations (VRM), probably carried by magnetite (or maghaemite), are also removed by thermal demagnetization (sample 245–01 in Fig. 2), but higher temperatures are frequently required (sample 48–01 in Fig. 2).

Generally one specimen from each core sample was measured. The core corrected NRM directions (Fig. 4) trend parallel to the present field (inclinations of the dipole field and the actual field are 47° and 42° respectively) resulting from secondary remanence directions carried by goethite. From the total number of 594 specimens stable ChRM directions were obtained from 560 specimens (Fig. 5) with approximately the same number of normal and reverse polarities. Mean values of normal ($decl. = 0° \pm 90°$) and reverse ($decl. = 180° \pm 90°$) polarities are $decl. = 359.0$, $incl. = 12.2$, ($\alpha_{95} = 3.7$, $k = 5.9$) and $decl. = 178.6$, $incl. = -23.2$, ($\alpha_{95} = 4.4$, $k = 4.9$) respectively. However a limited number of specimens show an unclear intermediate direction. We chose windows of $-40° < decl. < +40°$, $-20° < incl. < +60°$ (normal polarity) and $140° < decl. < 220°$, $-60° < incl. < +20°$ (reverse polarity) to facilitate the classification. These limitations reduce the number of accepted specimens to 436, 233 of normal polarity and 203 of reverse polarity. The corresponding mean directions are $decl. = 358.3$, $incl. = 15.5$ (normal polarity; $\alpha_{95} = 2.6$, $k = 14.1$) and $decl. = 178.2$, $incl. = -24.3$ (reverse polarity; $\alpha_{95} = 3.0$, $k = 12.0$). A considerable inclination flattening of about 20° is obvious, even when the northwards drift of the Indian Plate since the Miocene with about 5 cm yr^{-1} is taken into account (e.g. Patriat & Achache 1984). A similar inclination error has also been reported from Siwalik sections in Pakistan and India by other authors (Tauxe & Opdyke 1982; Johnson *et al.* 1983).

The fabric of the sandstones has been investigated by studies of the anisotropy of magnetic susceptibility (AMS, see e.g. Hrouda 1982). From NRM and susceptibility (x) data shown in figure 3, it is evident that x is dominated by paramagnetic minerals below 1800 m (no correlation between x and NRM: x homogeneous around 10^{-4} SI, NRM varying by a factor 100), whereas ferrimagnetic minerals (probably magnetite) control x above 1800 m (correlation between x and NRM, both varying by a factor 100). The shape of the paramagnetic minerals and of the magnetite particles is oblate shown by the factor $E = x_2^2/(x_1 x_3)$ (Hrouda 1982) generally larger than 1 (x_1, x_2, x_3 are the maximum, intermediate, minimum axes of the AMS tensor respectively). The factor $P = x_1/x_3$ (Nagata 1961) is mainly around 1.03 and indicates that the degree of anisotropy is relatively low for these sandstones. The data prove a sedimentary fabric independent of a para- or ferrimagnetic

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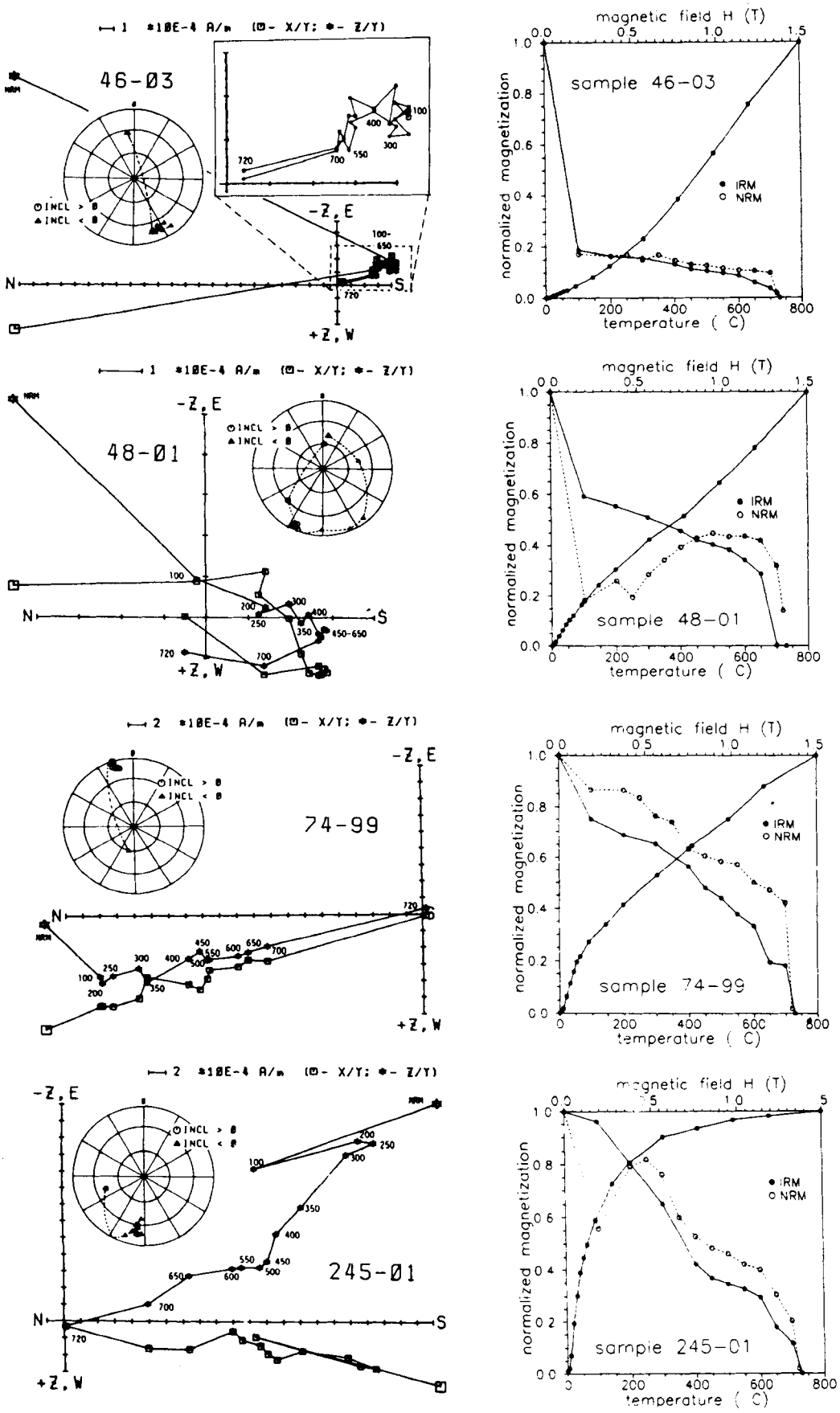


Figure 2. Orthogonal vector plots, equal area projections (bedding corrected) for the thermal demagnetization of typical samples. Temperatures are given in °C. IRM acquisition curves and thermal demagnetization of IRM and NRM are also shown.

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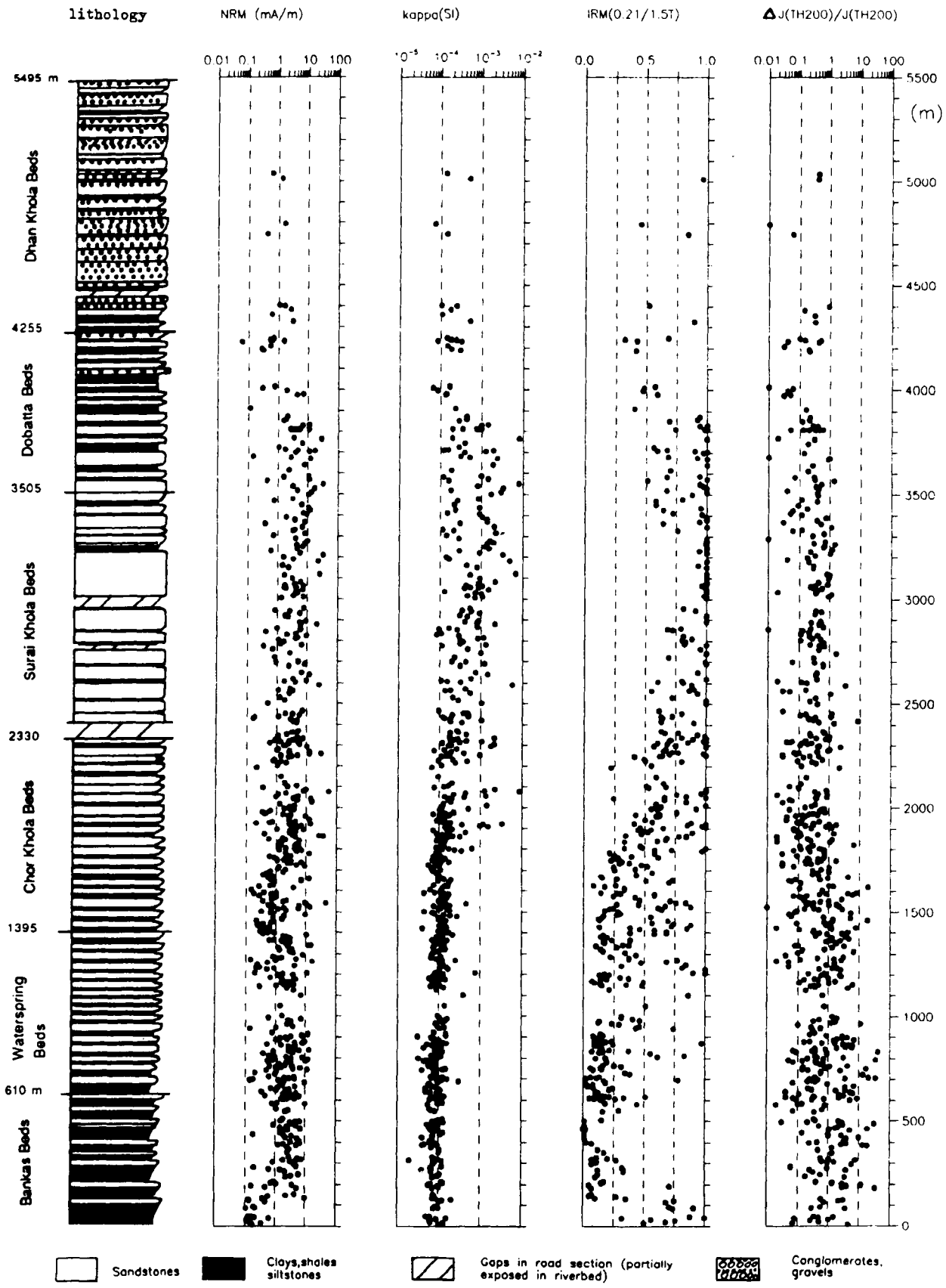


Figure 3. Lithology and rock magnetic properties throughout the Surai Khola section. NRM: natural remanent magnetization; kappa: susceptibility; IRM(0.21 T/1.5 T): ratio of IRM's acquired in fields of 0.21 and 1.5 T; dJ(TH200): intensity of the remanence demagnetized from 0° to 200 °C, J(TH200): residual remanence intensity at 200 °C.

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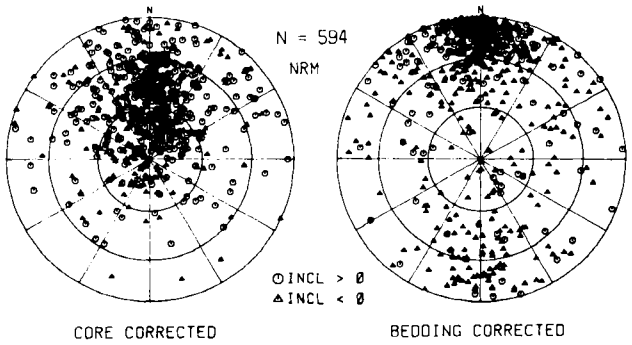


Figure 4. Equal area projection of core corrected and bedding corrected NRM directions from all sandstone specimens.

control of x: the maximum axes are grouped within the bedding plane, whereas the minimum axes are perpendicular to it. The preferential direction of the maximum axes is approximately perpendicular to the mean ChRM direction. Hence a substantial overprint of ChRM directions by alignment of ferrimagnetic grains seems unlikely.

MAGNETOSTRATIGRAPHY AND SEDIMENT ACCUMULATION RATES

The polarity sequence has been established with the 436 specimens from the windows defined above. Declinations are taken as the main parameter (around 0° and 180° for normal and reverse polarities respectively). Inclinations are less critical (about +20° and −20° for normal and reverse polarities respectively) but support the decision. Individual polarity intervals are only accepted if they are established by at least two specimens. Polarity transitions are fixed between neighbouring sample positions. The final result is shown in Fig. 6. The polarity sequence is well established throughout the section except in the lowest 100 m and above about 3900 m where too few specimens are available.

In Fig. 7 the final polarity sequence is compared with the standard polarity time-scale of Harland *et al.* (1982). First clues for the correlation result from the discovery of Gomphoterium sp. at 350 m corresponding to *Chinji* (about 10–13 Myr) and other fossils above 3000 m corresponding to *Tatrot/Pinjore* (< about 5 Myr) (Corvinus 1988). With a high probability the long normal polarity interval between about 800–1400 m correlates with chron 5. An alternative

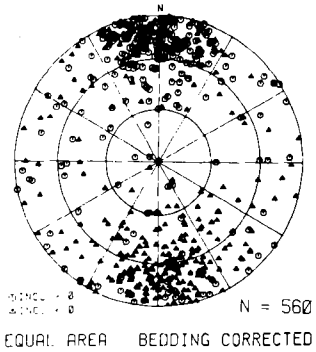


Figure 5. Equal area projection of the bedding corrected characteristic remanence directions from specimens.

correlation seems not to be possible with regard to Gomphoterium sp. below this interval. Further correlations are based on the comparison of polarity patterns. The entire section is found to range from chron 1r or chron 2r (Matuyama) to probably chron 5AB (1–2 Myr to about 13 Myr).

In Fig. 8 sediment accumulation rates are determined and compared with other Siwalik data from the literature. For the period between the top of the section and chron 5 (10.5 Myr) the sediment accumulation rate is approximately constant [42 cm $(10^3\ \mathrm{yr})^{-1}$]; below chron 5 it decreases to 17–31 cm $(10^3\ \mathrm{yr})^{-1}$. Previously published values range from 14–52 cm $(10^3\ \mathrm{yr})^{-1}$ and 13–23 cm $(10^3\ \mathrm{yr})^{-1}$ respectively. Extrapolation to a sediment accumulation rate of zero gives an age of about 15 Myr, probably corresponding to the onset of the uplift of the Himalayan mountain range.

The Surai Khola section certainly represents a complete record without any major gaps. Due to the rise of the Himalaya continuous sedimentation is expected whereas erosion should have been of minor importance. The frequency of reversals in the Surai Khola section equals the frequency of reversals in the standard polarity time-scale. Based on the sediment accumulation rates, erosion in the order of 100 m is required to 'loose' a reversal. Furthermore there is no geological evidence for the existence of any important sedimentation gaps.

Further magnetostratigraphic studies in the Nepalese Siwaliks are in progress and will be extended to different areas.

ACKNOWLEDGMENTS

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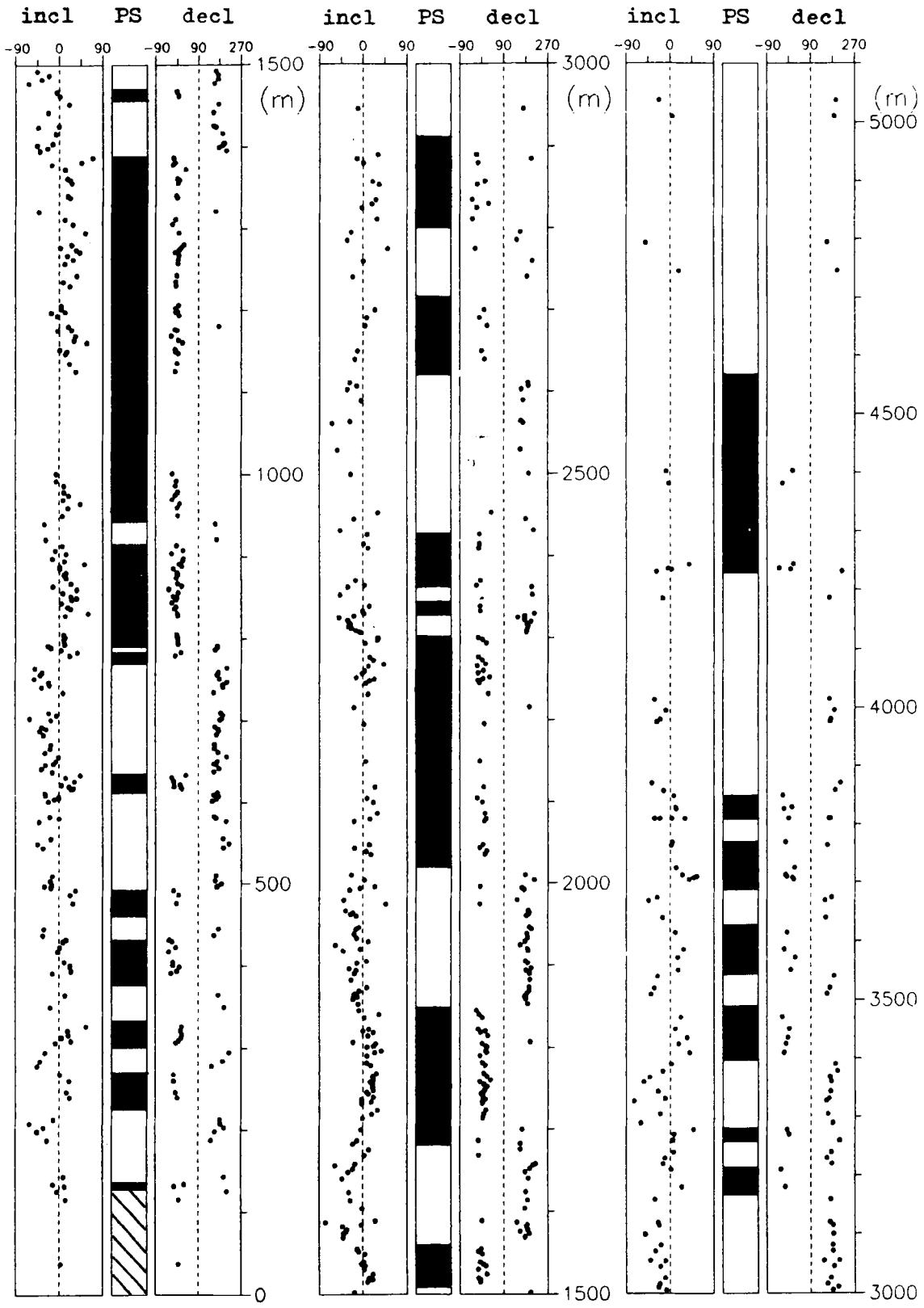


Figure 6. Declinations, inclinations and inferred polarity sequence throughout the Surai Khola section. Black: normal polarity, white: reverse polarity.

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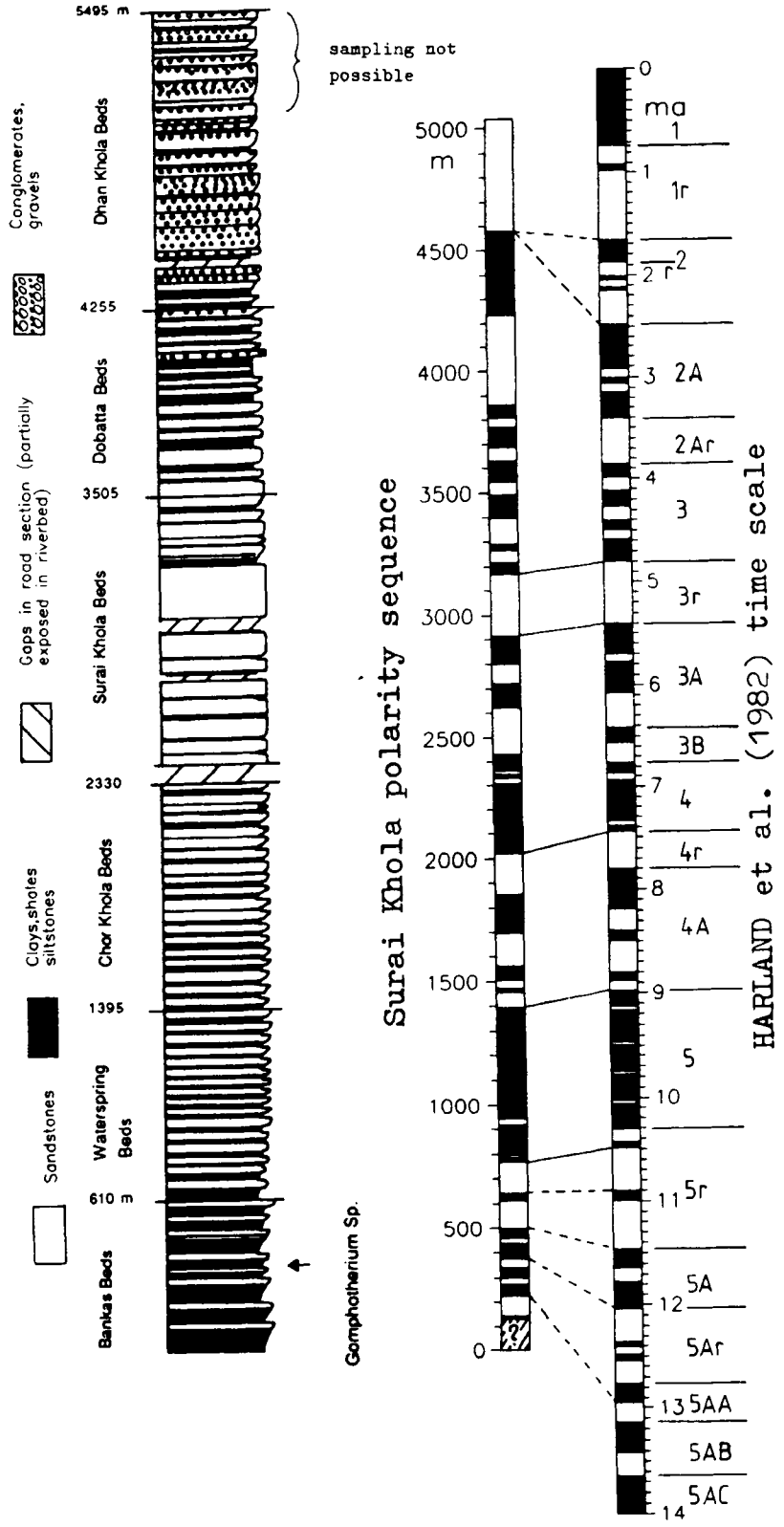


Figure 7. Correlation of the Surai Khola polarity sequence with the standard polarity time-scale of Harland *et al.* (1982). Full lines represent reliable correlations, more uncertain correlations are shown by broken lines. Black: normal polarity, white: reverse polarity.

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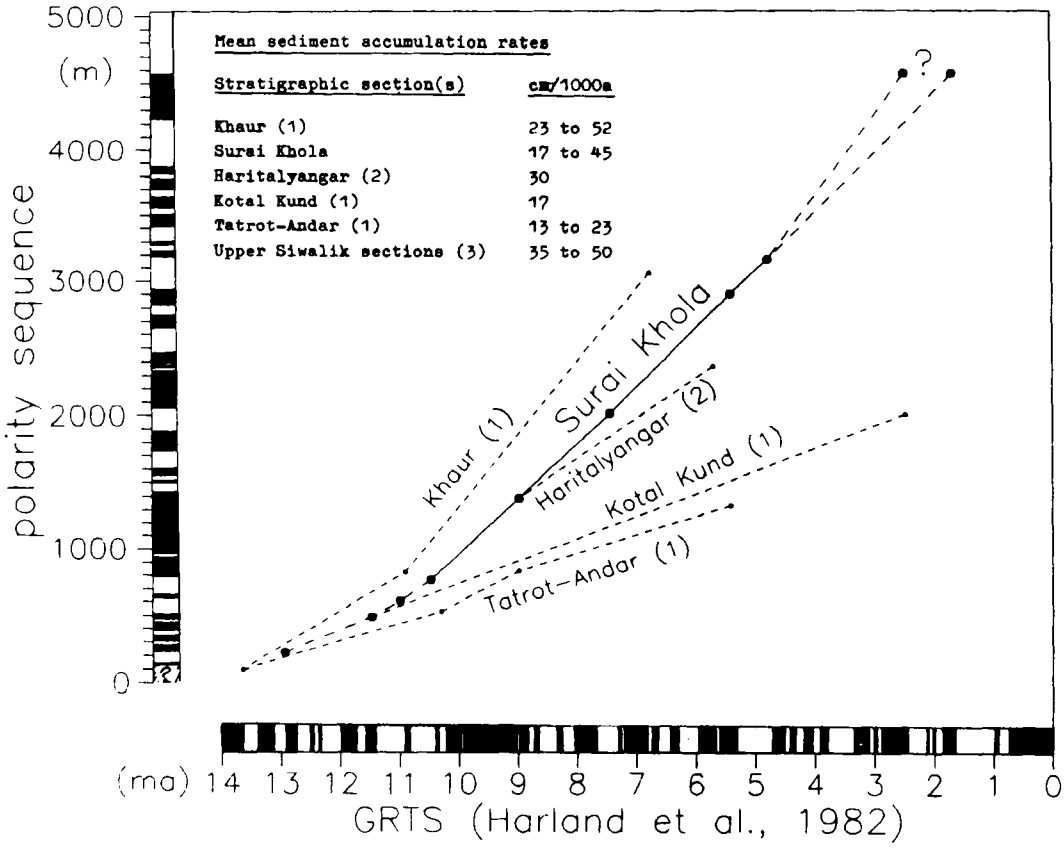


Figure 8. Sediment accumulation rates of the Surai Khola section deduced from the polarity sequence plotted versus a standard polarity time-scale. Results from literature are shown within the determined age range starting at the lower age limit on the Surai Khola curve. (1) Johnson *et al.* (1982), (2) Johnson *et al.* (1983), (3) Opdyke *et al.* (1979).

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